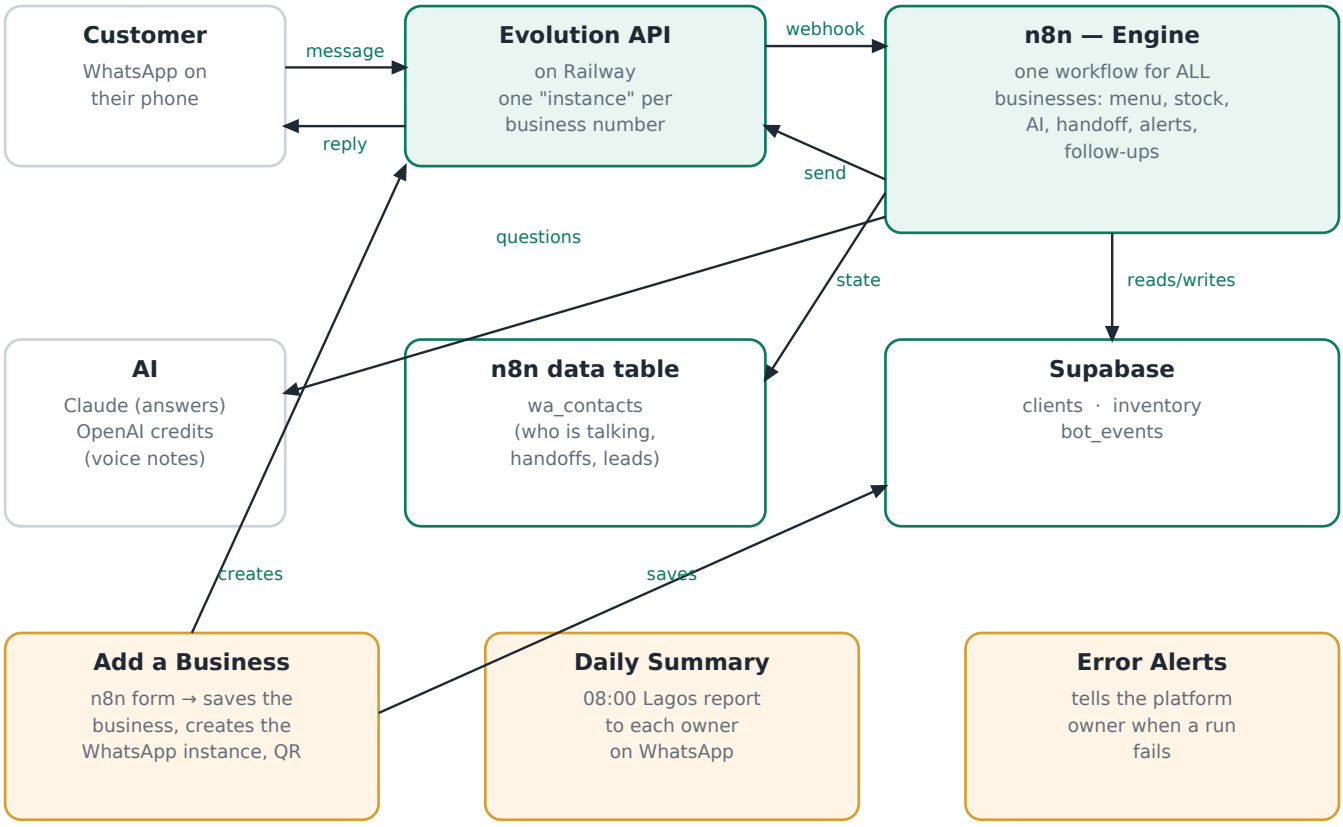
WhatsApp Bot — Configuration Map

How the multi-business WhatsApp assistant is wired, what each piece does, how to add a new business, and how to move everything to a new n8n with one click.

The 30-second version

- **One engine, many businesses.** A single n8n workflow answers WhatsApp for every business. Nothing about a business lives in the workflow — it all lives in that business's row in the Supabase **clients** table and its products in **inventory**.
- **Each business = one WhatsApp number = one Evolution "instance".** The instance name is how the engine knows which business a message is for.
- **Adding a business** = fill the **Add a Business** form and scan a QR code. Or do the same three things by hand (page 7): add a clients row, add products, create the instance with the engine webhook.
- **Changing a business** (prices, products, hours, tone, on/off) = the **Edit a Business** form: type its name, change the pre-filled page, press Save changes. Applies to the very next message.

System map



A customer message travels: phone → Evolution (the WhatsApp connection) → the n8n engine webhook → the engine looks up the business, the customer's state and (if needed) asks Claude → the reply goes back out through Evolution. The three yellow workflows run alongside it.

1 · Accounts, addresses and keys

Everything the bot depends on. Keys are never written in this document — the "where to find it" column tells you where each lives.

Service	What it is used for	Address / id	Where the key lives
n8n Cloud	Runs the 4 workflows	https://dynamit3.app.n8n.cloud	Your n8n login. Credentials are stored inside n8n.
Evolution API (Railway)	The WhatsApp connection for every business number	https://evolution-api-production-4056.up.railway.app Manager: https://evolution-api-production-4056.up.railway.app/manager	Railway → project **thriving-peace** → service **evolution-api** → Variables → **AUTHENTICATION_API_KEY** (the global key)
Supabase	Business settings, products, alert log	Project ref **esuatccbicekcohzgcvd**	n8n credential **Supabase account** (OAuth sign-in)
Anthropic (Claude)	Writes the AI answers	Model: Claude Sonnet	n8n credential **Anthropic account** (API key from console.anthropic.com)
OpenAI via n8n credits	Turns voice notes into text	—	n8n "Gateway credits" — built in, no key
Stripe / Paystack	Payment links customers are sent	Per product: inventory.payment_url	Your Stripe/Paystack dashboard
Gmail (optional)	Email copies of alerts	—	n8n credential (Sign in with Google). Email Alert step is off until connected.

Credentials inside n8n

n8n credential	Type	How it must be set up	Used by
Header Auth account	Header Auth	Name: **apikey** · Value: the Evolution **global key** (AUTHENTICATION_API_KEY)	Every call to Evolution: send message, fetch photo/voice, create instance, follow-ups, daily summary
Supabase account	Supabase OAuth2	Sign in to Supabase and allow access to the project	Fetch Client Config, stock steps, Log Event, live_services, Add a Business, Daily Summary
Anthropic account	Anthropic	Paste an Anthropic API key	The Claude step under AI Assistant

Important addresses

What	Address
Engine webhook (Evolution sends every message here)	https://dynamit3.app.n8n.cloud/webhook/faa757c3-e220-420d-bd89-803d5393f58e
Add a Business form (n8n login needed)	https://dynamit3.app.n8n.cloud/form/ca2ecfe7-8d12-4a43-ba77-14a218798100
Evolution manager (instances, QR codes, webhooks)	https://evolution-api-production-4056.up.railway.app/manager

2 · The engine, stage by stage

Workflow: **WhatsApp Bot — Engine (all businesses)**. Two triggers: the **Webhook** (every WhatsApp message) and **Every Hour** (follow-ups). Read top to bottom; each row is a step you will see on the n8n canvas.

Stage	Steps (n8n nodes)	What happens
1 · Receive	Webhook → Parse Command → Settings	Settings reads the Evolution address and Supabase project from the wa_settings table. Keeps only real new messages (event messages.upsert). Drops groups, statuses, the bot's own replies. Works out the customer number, text, photo or voice note. A message typed by your team on the business phone is marked as "agent".
2 · Which business?	Fetch Client Config → Client Active? → Business Profile	Finds the clients row whose **instance_name** matches the Evolution instance. Stops if none, or **active** is false. Business Profile builds the menu (yours or the default) and, if **ai_prompt** is empty, writes AI instructions from about / faqs / tone.
3 · Who is talking?	Get Contact → Decide → Save Contact → Should Reply?	Loads this customer's state from **wa_contacts** (key = instance:phone). Decide picks the command: menu, menu option 1–5, stock desk, AI question, photo, voice note. Handles handoff to a person, #close, the 8-messages-a-minute spam limit. State is saved every time; if the bot should stay quiet, it stops here.
4 · Route	Route Command	Stock → Get Item (Stock). Sell / Restock → Get Item (Change) → Calculate New Quantity → Can Update? → Update Quantity. Report (list, low, services) → Get All Items. Ask → Has Voice? / Has Photo? → AI. Everything else → Compose Reply.
5 · AI answers	Has Voice? → Fetch Voice → Voice File → Transcribe Voice → Use Transcript; Has Photo? → Fetch Image → Photo File; AI Assistant (+ Claude, Conversation Memory, live_services)	Voice notes become text first. The AI sees the business instructions, the customer's name, recent chat, and can look up live products, prices and payment links (live_services). It adds a hidden [[LEAD]] tag when someone wants to buy or needs the team.
6 · Write the reply	Compose Reply	Builds every WhatsApp message: welcome menu, About, products list with prices and Pay links, stock desk, sale receipts, "connecting you to our team", chat closed. Also writes the owner alert text (lead, person requested, sale, couldn't answer).
7 · Alerts & logging	New Contact Alert → Expand Recipients → Loop Over Recipients → Send WhatsApp Message → Wait Between Sends; Has Alert? → Log Event / Email Alert; Is Lead? → Mark Lead	Adds a "New contact" alert for first-time customers. Sends the reply to the customer and the alert to every **alert_numbers** phone, 1.5 s apart. Logs the alert to **bot_events** (feeds the daily report), emails it (when Gmail is on) and marks leads for follow-up.
8 · Follow-ups (hourly)	Every Hour → Get Clients → Recent Leads → Due Follow-ups → Send Follow-up → Mark Followed	Leads and new contacts who went quiet 24–48 hours ago get one friendly check-in from their business — never if a person is handling the chat.

The three companion workflows

Workflow	Trigger	What it does
WhatsApp Bot — Add a Business	Form (n8n login)	Prepare Business → Create Client → Any Services? → Split Services → Create Products → Create WhatsApp → Scan to Connect (shows the QR).

Workflow	Trigger	What it does
WhatsApp Bot — Daily Summary	08:00 Africa/Lagos	Get Clients → Last 24h Events → Build Summary → Send Summary. One report per business with daily_summary = true.
WhatsApp Bot — Error Alerts	Any engine failure	Compose Alert → Settings → WhatsApp the Owner. Set as the Error Workflow of the Engine and Daily Summary.
WhatsApp Bot — Edit a Business	Form (n8n login)	Find a business → pre-filled page with its details and products → Save changes. Add a product line to add it, delete a line to remove it.
WhatsApp Bot — Setup / Move	Form (n8n login)	One-click setup on any n8n: creates wa_settings and wa_contacts, saves the settings, points every business's WhatsApp at this n8n (page 8).

3 · How the bot behaves (cheat sheet)

Customer sends	Bot does
hi, hello, menu, help, 0, start…	Welcome + menu (options 1–4 from the business, 5 = Talk to a person).
1 – 4 (also "option 3", "#3", or the 3 emoji)	Runs that menu option: about = shows clients.about; services = products list; ask = asks the AI the option's question.
5, "talk to a person", "human", "agent"	Says the team is coming, alerts the owner, and goes **silent** in that chat.
Any question / sentence	AI answers using the business info and live products. Hot leads alert the owner.
A photo	AI looks at it (with any caption) and answers.
A voice note	Transcribed, then answered like a typed message.
More than 8 messages in a minute	Ignored (spam / cost protection).

Owner / staff (numbers in alert_numbers) send	Bot does
6, staff, desk	Stock desk menu
list · low	All stock · items with 5 or fewer left
stock rice	Availability and price of one item
sell rice 2 · restock rice 10	Updates the quantity, confirms, alerts the other owner numbers (with a low-stock warning)

Handing a chat to a person

- The bot goes quiet when the customer picks 5 **or** the moment anyone replies from the business phone in that chat.
- It comes back when: the team sends **#close** (instant, customer is told), nobody writes for **1 hour** after the team's last reply, nobody from the team replies within **24 hours**, or the customer types **menu**.

Alerts the owner receives (WhatsApp, and email when Gmail is on)

New contact · **New lead** · **Person requested** · **Assistant couldn't answer** · **New sale / Restock** (with "only X left"). Every alert is also stored in **bot_events** and counted in the 08:00 daily report.

4 · What you edit per business

Quick way: the **Edit a Business** form — type the business name, change the pre-filled page (products one per line: Name | price | stock | payment link | description; add a line to add, delete a line to remove), press **Save changes**. Or edit in Supabase → Table Editor. Edits apply on the next message.

clients — one row per business

Column	Required	What it does
client_id	yes	Unique id (e.g. mama_tee_bakery). Links the business to its products and events.
instance_name	yes	Must equal the Evolution instance name of its WhatsApp number exactly.
business_name	yes	Shown in the welcome, alerts and AI.
alert_numbers	yes	Owner/staff numbers, with country code, no +. Get alerts, the daily report and the stock desk. Format: {2348012345678,2348098765432}
about	recommended	What the business does — the "About" option and the AI's main knowledge.
faqs	recommended	Hours, location, delivery fees, how to pay (bank details), policies. The AI answers from this.
tone	optional	How the AI should sound, e.g. "Friendly Nigerian English; reply in Pidgin if they do."
tagline · website	optional	Shown under the welcome / in the footer.
currency	default NGN	NGN, GHS, KES, ZAR, USD, GBP, EUR… used to format prices.
menu	optional	Up to 4 options (JSON, below). Empty = default menu: About · Products & prices · How to order & pay · Hours & location.
ai_prompt	optional	Full custom AI instructions. Empty = built automatically from about + faqs + tone (recommended).
services_note	optional	Extra line under the products list (e.g. "Custom cakes — quoted on request").
alert_email	optional	Where email alerts go (once Gmail is connected).
daily_summary	default true	false = no 08:00 report for this business.
active	default true	false = bot stops answering for this business (nothing is deleted).

Example custom menu (paste into the menu column):

```
[
 {"emoji": "*", "label": "Cakes & prices", "action": "services"},
 {"emoji": "*", "label": "Delivery areas", "action": "ask",
  "question": "Where do you deliver and how much is it?"},
 {"emoji": "*", "label": "How to pay", "action": "ask",
  "question": "How can I pay?"},
 {"emoji": "*", "label": "About us", "action": "about"}
]
(Replace each * with any emoji you like.)
```

inventory — one row per product or service

Column	What it does
client_id	Which business it belongs to (same as clients.client_id).
item_name	Product name (unique per business).
price	Number only, no symbol — e.g. 25000.
quantity	Stock. 0 shows as unavailable/sold out. Updated by sell / restock.
description	One line shown under the product.

Column	What it does
payment_url	Optional checkout link (Stripe Payment Link, Paystack page…). Shown as "Pay:" and sent by the AI when someone wants to buy.

Read-only (filled by the bot): **bot_events** in Supabase (every alert) and the n8n data table **wa_contacts** (one row per business + customer: session, handoff times, lead / follow-up times, spam counter).

5 · Add a new business

Before you start, have: the business's WhatsApp phone in hand (a number not already linked to another bot), the owner's number(s), and a short description, FAQs and product list.

Option A — the form (2 minutes)

1. Log in to n8n, then open the form: **https://dynamit3.app.n8n.cloud/form/ca2ecfe7-8d12-4a43-ba77-14a218798100**
2. Fill in name, alert numbers, about, FAQs, products (one per line: **Name | price | stock | payment link | short description**), tone and currency. Submit.
3. A page shows a QR code. On the business phone: WhatsApp → Settings → **Linked devices** → **Link a device** → scan.
4. Send "hi" to the business number from another phone. You should get its welcome menu, and the owner gets a "New contact" alert.

If the QR expired: open the Evolution manager, find the instance (its name is shown on the result page), press **Get QR code** and scan again. If the page says "WhatsApp not created", the business is saved — finish with steps B3–B4 below.

Option B — by hand (no form, no help needed)

B1. Add the business — Supabase → SQL Editor → paste, change the values, Run:

```
insert into public.clients
  (client_id, instance_name, business_name, alert_numbers, about, faqs, tone, currency, tagline)
values
  ('mama_tee_bakery', 'mama_tee_bakery', 'Mama Tee Bakery',
   '{2348012345678}',
   'Fresh bread, cakes and small chops in Lekki, Lagos.',
   'Open Mon-Sat 8am-7pm. Delivery in Lekki N2,000. Pay by transfer: GTBank 0123456789.',
   'Friendly Nigerian English; reply in Pidgin if they do.',
   'NGN', 'Fresh bread & cakes, delivered in Lekki');
```

B2. Add products:

```
insert into public.inventory (client_id, item_name, price, quantity, description, payment_url) values
  ('mama_tee_bakery', 'Birthday cake 8 inch', 25000, 10, 'Vanilla or chocolate', null),
  ('mama_tee_bakery', 'Small chops tray', 15000, 20, '50 pieces', 'https://paystack.com/pay/xyz');
```

B3. Create the WhatsApp instance — in the Evolution manager:

1. Open **https://evolution-api-production-4056.up.railway.app/manager** and log in with the global key.
2. **Instance +** → Name: exactly the **instance_name** from B1 (e.g. mama_tee_bakery) → Channel: **Baileys** → Create.
3. Open the new instance → **Events → Webhook**: Enabled **on** · URL: the engine webhook below · Webhook by Events **off** · Webhook Base64 **off** · events: **MESSAGES_UPSERT only** → **Save** (bottom of the page).

```
https://dynamit3.app.n8n.cloud/webhook/faa757c3-e220-420d-bd89-803d5393f58e
```

B4. Connect the phone: on the instance's dashboard press **Get QR code** and scan it from the business phone (Linked devices → Link a device). Status should turn **Connected / open**.

B5. Test: message "hi" from another phone → menu arrives; owner gets "New contact". In n8n → the engine → **Executions** you will see the run.

6 · Move to a new n8n — one-click setup

Use this when you switch n8n accounts (for example when credits run out). Businesses, products and WhatsApp numbers stay in Supabase and Evolution — only n8n moves. About 10 minutes. Nothing install-specific is written into the workflows: the Evolution address, Supabase project, engine address and your alert number live in one n8n data table, **wa_settings**, which the Setup / Move form fills in.

Step 1 — create the three logins (n8n → Credentials → Create), with exactly these names

Name	Type	Fill in
Header Auth account	Header Auth	Name: **apikey** · Value: Evolution global key (Railway → evolution-api → Variables → AUTHENTICATION_API_KEY)
Supabase account	Supabase OAuth2 API	Sign in to Supabase and allow the project
Anthropic account	Anthropic	An Anthropic API key

Step 2 — import the six files from the repo folder n8n/whatsapp/ (Workflows → Import from file)

- 1-setup-move.workflow.json
- 2-engine.workflow.json
- 3-add-a-business.workflow.json
- 4-daily-summary.workflow.json
- 5-error-alerts.workflow.json
- 6-edit-a-business.workflow.json

Logins attach by name. If a step still shows a red credential warning, open it and pick the login with the matching name.

Step 3 — click the button

Open **WhatsApp Bot — Setup / Move**, publish it and open its form. Check the pre-filled values, leave "Engine webhook address" empty and press **Set up**. In one go it creates the wa_settings and wa_contacts tables, saves the settings every workflow reads, and points **every business's WhatsApp** at this n8n. The result page shows each business as done or failed.

Step 4 — publish and finish

1. Publish Engine, Add a Business, Edit a Business, Daily Summary and Error Alerts.
2. In the Engine and Daily Summary: Workflow settings → Error workflow → **WhatsApp Bot — Error Alerts**.
3. Send "hi" to any business number.

Safe to run Setup / Move again at any time (e.g. after the Evolution address changes). On a **self-hosted** n8n, add an OpenAI login to the Transcribe Voice step (voice notes use n8n Cloud's built-in credits otherwise). Chats with a person and recent AI memory start fresh after a move; business data is untouched.

Moving Supabase or Evolution too (rare)

- **New Supabase project:** run n8n/whatsapp/schema.sql in its SQL Editor, copy your clients and inventory rows across, and enter the new project ref in Setup / Move.

- **New Evolution server:** deploy the Evolution API template on Railway with a strong AUTHENTICATION_API_KEY, put the new address in Setup / Move, then re-scan each business's WhatsApp (Get QR code in the manager).

7 · Troubleshooting

Symptom	Check this
Bot does not reply at all	1) Evolution manager: instance **Connected**? 2) Its webhook URL is the engine URL, MESSAGES_UPSERT on, Webhook by Events off, saved? 3) n8n engine is **published**? 4) clients row: instance_name matches exactly and active = true? 5) n8n → Executions: did a run start?
Run starts but stops early	Executions → open the run. Stopped at Fetch Client Config / Client Active? = no matching clients row. Stopped at Should Reply? = chat is with a person (send #close) or spam limit.
Reply fails with 401	Header Auth credential: Name must be **apikey**, Value the current global key.
"Sorry, I couldn't answer that just now"	Anthropic key out of credit or invalid — check the Claude step's error in the execution.
Voice notes not understood	n8n gateway credits used up — add an OpenAI credential to the Transcribe Voice step.
Bot keeps talking after option 5	Should not happen any more; if a customer typed something else, reply from the phone (bot goes quiet) and use #close when done.
Wrong prices / products	Edit the inventory rows for that client_id.
No 08:00 report	Daily Summary published? clients.daily_summary true? alert_numbers set?

8 · Running 250 businesses

- **One WhatsApp number per business**, each its own Evolution instance. The engine needs no changes.
- **Evolution server:** each connected number uses memory — scale the Railway service up as you add businesses and watch its metrics.
- **n8n plan:** every message is one execution (plus one per hour for follow-ups). Pick a plan with enough monthly executions and concurrent runs for your busiest hour.
- **AI cost:** menu taps, stock commands and handoffs use no AI; only free-text questions, photos and voice notes do. Budget per business accordingly and price the subscription above it.
- **WhatsApp rules:** unofficial (Baileys) connections can be banned for spam. Never use the bot for bulk marketing; keep follow-ups to the one built-in nudge.